

GPU-Accelerated TMCMC with Vectorised Random Walk Mutation

Keisuke Nishioka

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1 Introduction

Transitional Markov Chain Monte Carlo [?] is a powerful Sequential Monte Carlo (SMC) method for Bayesian model updating, but its computational cost scales with the number of particles N_p and the cost of each likelihood evaluation. For the 5-species biofilm Hamilton ODE system (20 parameters, stiff implicit Euler solver), a single TMCMC run with $N_p = 500$ particles requires 6–10 hours on a 12-core CPU cluster.

Recent advances in hardware-accelerated probabilistic programming [?, ?, ?] suggest that `jax.vmap` can vectorise independent MCMC chains across GPU cores, offering order-of-magnitude speedups. However, the interaction between GPU vectorisation and different MCMC mutation kernels is non-trivial: gradient-based samplers (NUTS, HMC) exhibit variable-length control flow that defeats SIMD parallelism [?].

This section presents our GPU TMCMC implementation in JAX, achieving a **200× speedup** over the CPU baseline by combining `vmap`-parallelised random walk (RW) mutation with a pure-JAX implicit ODE solver. We document the failure modes of NUTS/HMC on GPU and provide architectural guidelines for GPU-friendly SMC.

2 Background: GPU-Friendly MCMC

2.1 SIMD Constraint and Mutation Kernels

GPUs execute instructions in SIMD (Single Instruction, Multiple Data) fashion: all threads in a warp run the same instruction on different data. This creates a fundamental distinction between mutation kernels:

Random Walk Metropolis–Hastings (RWMH). Each particle undergoes a fixed-cost operation: propose $\theta^* = \theta + \varepsilon$, evaluate $\log \mathcal{L}(\theta^*)$, accept/reject. All particles execute identical operations with different inputs—*perfect* for `vmap`.

NUTS (No-U-Turn Sampler). Each chain adaptively chooses a different number of leapfrog steps via the U-turn criterion [?]. With `vmap`, all chains must execute $\max_i(\text{depth}_i)$ iterations; if 99/100 chains need depth 3 but one needs depth 10, then 99% of GPU compute is wasted for 7 iterations. [?] show this can waste up to 90% of theoretical throughput.

HMC with fixed trajectory. ChEES-HMC [?] uses a fixed trajectory length to avoid the variable-depth problem, but still requires sequential gradient evaluations within each leapfrog trajectory—limiting the parallelism to the particle dimension only.

2.2 Related Work

Table ?? summarises existing GPU SMC/MCMC implementations.

Table 1: Existing GPU-accelerated SMC/MCMC implementations.

Reference	Method	Speedup	Notes
[?]	TFP MCMC on GPU	—	Engineering principles for GPU MCMC
[?]	ChEES-HMC	$\sim 5\times$	Fixed trajectory length for GPU
[?]	BlackJAX SMC	$\sim 10\text{--}40\times$	Composable JAX SMC framework
[?]	FSM-NUTS	up to $10\times$	Eliminates vmap synchronisation waste
[?]	Survey	—	NUTS $43\times$ slower with vmap vs pmap
[?]	Parallel resampling	$10\text{--}95\times$	Metropolis resampling on GPU
[?]	Annealed flow transport	$\sim 20\times$	NF + SMC on GPU
This work	RW TMCMC, vmap	$\sim 200\times$	500 particles, biofilm ODE

3 Architecture

3.1 System Overview

The GPU TMCMC pipeline consists of three components:

1. **Pure-JAX ODE solver** (`hamilton_ode_jax.py`): 5-species Hamilton system solved via implicit Euler with custom Gaussian elimination (partial pivoting, 12 Newton iterations per step). All operations remain on GPU VRAM—no host $\leftrightarrow$ device transfers during ODE integration.
2. **Vectorised likelihood**: `jax.jit(jax.vmap(log_likelihood))` maps a single-particle log-likelihood function over all N_p particles simultaneously.
3. **TMCMC engine** (`tmcmc_nuts_engine.py`): ESS-based temperature scheduling, systematic resampling, and RW/HMC/NUTS mutation—with RW as the GPU-optimised default.

3.2 vmap Random Walk Mutation

Algorithm ?? describes the vectorised RW mutation kernel.

Algorithm 1 vmap Random Walk mutation for GPU TMCMC

Require: Particles $\{\theta_i\}_{i=1}^{N_p}$, temperature β_j , weighted covariance $\hat{\Sigma}$, number of mutation steps M

Ensure: Mutated particles $\{\theta'_i\}_{i=1}^{N_p}$

- 1: `logL_vmap` $\leftarrow$ `jax.jit(jax.vmap(log L))` {Compile once, reuse}
 - 2: $\Sigma_{\text{prop}} \leftarrow c^2 \hat{\Sigma}$ where $c = 2.38/\sqrt{d}$ {Optimal RW scaling [Roberts et al., 1997]}
 - 3: **for** $m = 1, \dots, M$ **do**
 - 4: $\varepsilon_i \sim \mathcal{N}(0, \Sigma_{\text{prop}})$ for all i {Vectorised proposal}
 - 5: $\theta_i^* \leftarrow \theta_i + \varepsilon_i$
 - 6: Clip θ_i^* to prior bounds $[\theta_{\min}, \theta_{\max}]$
 - 7: $\{\log \mathcal{L}(\theta_i^*)\}_{i=1}^{N_p} \leftarrow \text{logL_vmap}(\{\theta_i^*\})$ {Single GPU call}
 - 8: $\log \alpha_i \leftarrow \beta_j (\log \mathcal{L}(\theta_i^*) - \log \mathcal{L}(\theta_i))$
 - 9: Accept $\theta_i \leftarrow \theta_i^*$ if $\log u_i < \log \alpha_i$ {Vectorised MH step}
 - 10: **end for**
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The critical insight is line 7: a single `logL_vmap` call evaluates all 500 particles’ likelihoods in one GPU kernel launch. On CPU, this requires 500 sequential ODE solves (~ 42 batches of 12 cores); on GPU, all 500 solves execute simultaneously.

3.3 Weighted Covariance Estimation

Following [?], the proposal covariance $\hat{\Sigma}$ is estimated from importance-weighted particles at each TMCMC stage:

$$\hat{\Sigma}_j = \sum_{i=1}^{N_p} \tilde{w}_i (\theta_i - \bar{\theta}_w)(\theta_i - \bar{\theta}_w)^\top, \quad \tilde{w}_i = \frac{w_i}{\sum_k w_k}, \quad (1)$$

where $w_i = \mathcal{L}(\theta_i)^{\beta_{j+1} - \beta_j}$ are the unnormalised importance weights.

4 Why NUTS/HMC Fail on GPU

We attempted three GPU strategies for gradient-based mutation and all failed:

4.1 vmap NUTS

`jax.vmap(nuts_step)` with nested `lax.fori_loop` for tree doubling. JIT compilation exceeded 26 minutes without completing. Root cause: XLA cannot inline the dynamic tree-building procedure across $N_p = 500$ particles, producing a combinatorially large HLO graph.

4.2 Per-Particle JIT NUTS

Running `jit(nuts_step)` independently for each particle. JAX caches a separate CUDA graph per particle, requiring $500 \times \mathcal{O}(\text{graph size})$ GPU memory. Result: out-of-memory (OOM) after ~ 395 cached graphs on RTX 4090 (24 GB VRAM).

4.3 Command Buffer Optimisation

Disabling JAX’s command buffering (`XLA_FLAGS=--xla_gpu_enable_command_buffer=false`) to reduce memory. Process killed due to memory leak in graph construction.

Fundamental limitation. The mismatch is architectural: NUTS/HMC require *deep sequential* gradient computation within each particle (leapfrog trajectory), while GPUs excel at *wide parallel* computation across particles. RW mutation has depth 1 (single likelihood evaluation per proposal), making it optimally suited for GPU vectorisation.

This aligns with [?], who report that `vmap`-ed NUTS can be **43× slower** than independent CPU chains due to warp divergence and synchronisation overhead.

5 Benchmark Results

5.1 Experimental Setup

- **Model:** 5-species biofilm Hamilton ODE, 20 parameters
- **GPU:** NVIDIA RTX 4090, 24 GB VRAM
- **CPU baseline:** PBS cluster, 12 cores (Intel Xeon), `estimate_reduced_nishioka.py`
- **Particles:** $N_p = 500$, RW mutation ($M = 10$ steps)
- **Condition:** Dysbiotic/HOBIC (DH baseline)

5.2 Results

Table 2: GPU vs CPU TMCMC benchmark (DH baseline, $N_p = 500$).

Metric	GPU (RTX 4090)	CPU (12-core)
Total stages	9	9
Total time	146.7 s	$\sim 21\,600$ s
Per-stage time	~ 14 s	$\sim 2\,400$ s
Final β	1.0	1.0
Acceptance rate	0.52	0.52
Speedup	$\sim 200\times$	

The acceptance rate is identical between GPU and CPU, confirming that the `vmap` implementation produces statistically equivalent samples. The $200\times$ speedup arises from two sources:

1. **Parallel likelihood** ($\sim 170\times$): 500 ODE solves simultaneously vs. 42 batches of 12.
2. **Reduced overhead** ($\sim 1.2\times$): No Python-level particle loops, no host $\leftrightarrow$ device transfers during ODE integration.

5.3 Scaling

GPU TMCMC scales linearly with N_p up to the VRAM limit. On the RTX 4090, we estimate a maximum of $\sim 4\,000$ particles before VRAM exhaustion (each particle’s ODE state occupies ~ 6 MB including Jacobian storage).

6 Pure-JAX ODE Solver

A critical enabler for the $200\times$ speedup is the pure-JAX implicit Euler solver for the Hamilton ODE system. Key design decisions:

1. **Custom linear algebra**: Gaussian elimination with partial pivoting, avoiding `jax.numpy.linalg.solve` which calls `cuSOLVER` (inefficient for small 5×5 systems).
2. **12 Newton iterations**: Increased from 6 to handle stiff regimes (dysbiotic conditions with rapid species transitions).
3. **`lax.scan` for time stepping**: Compiles the entire time integration into a single fused GPU kernel—no Python-level loop overhead.
4. **Float64 throughout**: Ensures numerical stability for stiff ODE; enabled via `jax.config.update("jax_enable_x64", True)`.

7 Advanced Extensions

7.1 Normalizing Flow Proposals

We implement RealNVP normalizing flow proposals [?] trained per TMCMC stage on the re-sampled particle cloud. The flow proposal replaces the Gaussian RW:

$$\theta^* = f_\phi^{-1}(z), \quad z \sim \mathcal{N}(0, I), \quad (2)$$

with Metropolis–Hastings correction:

$$\log \alpha = \beta_j(\log \mathcal{L}(\theta^*) - \log \mathcal{L}(\theta)) + \log q_\phi(\theta) - \log q_\phi(\theta^*). \quad (3)$$

This increases the acceptance rate from $\sim 52\%$ (RW) to $\sim 95\%$ (flow), at the cost of flow training (~ 2 s per stage).

7.2 Waste-Free SMC

Following [?], we retain all M mutation proposals (not just the final accepted state), effectively increasing the effective sample size by a factor of ~ 2 for the same number of likelihood evaluations. Combined with GPU `vmap`, this enables high-quality posterior estimation in under 3 minutes.

8 Discussion

Our results demonstrate that **GPU TMCMC with RW mutation is the optimal configuration** for the biofilm inverse problem. The key insight is that the embarrassingly parallel structure of RW mutation—identical fixed-cost operations across particles—maps perfectly onto GPU SIMD architecture.

This finding has broader implications:

- For SMC methods, the theoretical superiority of gradient-based kernels (higher ESS per sample) is offset by their GPU incompatibility. The *effective samples per second* metric favours RW on GPU.
- The $200\times$ speedup makes real-time Bayesian model updating feasible: a full posterior can be obtained in <3 minutes, enabling interactive exploration of model parameters.
- For stiff ODE systems, a pure-JAX solver with custom linear algebra outperforms general-purpose GPU libraries (cuSOLVER) by avoiding kernel launch overhead for small systems.

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